Buy versus Rent in England: A Cash-Flow-Matched Horse Race for the First-Time Buyer, 2005–2026

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Abstract

I compare the financial outcomes of buying a first home and renting it while investing, using a monthly cash-flow-matched simulation on official English data from January 2005 to April 2026. A leveraged first-time buyer (10% deposit, 30-year mortgage at the Bank of England effective rate, period-accurate stamp duty) is matched against a renter who invests the deposit-equivalent and every monthly cost difference in a global equity index fund. To compare like with like, I construct a representative English dwelling whose price and rent share one population-weighted regional composition, avoiding the bias that arises from combining the transaction-weighted national house-price average with the tenancy-weighted national rent average. Across 85 quarterly entry cohorts the two strategies finish near parity — buying produced greater terminal wealth in 53% of cohorts — but with wide dispersion governed by entry conditions: the price-to-rent ratio and the mortgage rate at purchase jointly explain 74% of the variation in outcomes, and buying's advantage is concentrated in the affordable North and Midlands and in the cheap-credit decade after 2008. Leverage is essential to this parity: an unlevered (cash) buyer never beats the renter. A forward-looking break-even metric — the required appreciation rate — indicates that at April 2026 conditions average English prices need grow only about 1% per year over a decade for buying to match a diversified equity investor.

JEL classification: D14, G11, G51, R21, R31

Keywords: housing tenure choice; buy versus rent; first-time buyers; user cost of housing; household portfolios; leverage; stamp duty

1 Introduction

Whether to buy a first home or to continue renting is, for most households, the largest financial decision they will ever take. In England the decision carries unusual cultural weight: around 65% of households own their home, first-time buyers account for a record share of new mortgage lending, and homeownership is widely regarded as the financially prudent choice (Ministry of Housing, Communities and Local Government, 2025; Halifax / Lloyds Banking Group, 2025). Yet the popular debate is conducted largely in slogans — renting as "dead money" on one side, the superiority of financial-market returns on the other — and the empirical evidence for England is surprisingly thin. No study has measured, on actual monthly data, how a leveraged first-time buyer would have fared against an otherwise identical renter who invested the difference in a diversified portfolio.

This paper runs that race. I construct a monthly cash-flow-matched simulation — in the spirit of Beracha and Johnson (2012) for the United States — on official English data from January 2005 to April 2026. The buyer purchases a home with a 10% deposit and a 30-year mortgage, pays the Bank of England's effective rate on new advances (recast monthly in the baseline), maintenance calibrated to UK evidence, and period-accurate first-time-buyer stamp duty. The renter occupies the same dwelling at the market rent, starts with capital equal to the buyer's deposit and purchase costs, and invests that capital — plus every subsequent monthly difference between the owner's outlays and the rent — in a global equity index fund (MSCI ACWI, measured in sterling, net of withholding taxes and fees). At the end of the holding period I compare the owner's net housing equity, after selling costs, with the renter's portfolio.

Two measurement choices distinguish the exercise. First, the comparison is run for every quarterly entry date between January 2005 and January 2026 — 85 overlapping cohorts — so that the role of *timing* is observed rather than averaged away. Second, I show that the obvious national implementation — dividing the official England average house price by the official England average private rent — is biased, because the two aggregates weight regions differently: the house-price average reflects the transaction mix, while the rent average is dominated by London's large rental stock. The resulting national price-to-rent ratio (16.9 in April 2026) falls *below that of every English region* (17.5–22.2), which would flatter homeownership in any cost comparison. I instead build a representative English dwelling whose price and rent are the same population-weighted average of the nine regions, and I additionally run the race region by region.

Three findings emerge. First, over these two decades the race was **remarkably close**: buying produced greater terminal wealth in 45 of 85 cohorts (53%), the mean renter/owner wealth ratio was 1.06, and the January 2005 cohort — the longest race — ended with the renter ahead by 9%. This near-parity is itself striking, since it obtains over a sample in which global equities returned 10.2% per year in sterling terms while the representative English dwelling appreciated at only 3.2%. Leverage reconciles the two: the buyer's mortgage converts modest house-price growth into competitive equity returns, and an unlevered (cash) buyer loses to the renter in every cohort.

Second, **entry conditions govern the outcome**. The cohorts sort into three regimes — renting won 2005–2008, buying won almost every entry from late 2008 to 2020, renting has won since — and two variables observable at purchase, the price-to-rent ratio and the mortgage rate, jointly explain 74% of the cross-cohort variation in the wealth ratio. Geography matters through the same channel: buying won a clear majority of cohorts in the North and Midlands, where price-to-rent ratios are low, and a small minority in the South and East, where they are high.

Third, a forward-looking **required appreciation rate** (RAR) — the annual house-price growth at which buying exactly breaks even with renting-and-investing, the user-cost equilibrium condition of Poterba (1984) and Himmelberg, Mayer and Sinai (2005) solved for the capital-gains term — indicates that conditions at the end of the sample mildly favour buying: at April 2026 prices, rents and rates, the representative dwelling needs to appreciate only about 1.0% per year over a decade to match a global equity investor earning a conservative 5.9% nominal, and over historical fixed ten-year holds buying was ex-post the better call in 88% of entry months.

The paper makes four contributions. (i) It provides the first monthly, cash-flow-matched buy-versus-rent backtest on UK data with an equity-investing renter, updating and substantially extending the annual, savings-account-counterfactual analysis of Mostafa and Jones (2019), whose sample ends in 2012. (ii) It documents and corrects the composition mismatch between the official national house-price and rent aggregates — a practical warning for any user of price-to-rent ratios built from UK HPI and ONS rent statistics. (iii) It introduces the required appreciation rate as a compact, forward-looking summary of buy-versus-rent conditions, and reports it historically, currently, and regionally. (iv) All code and data are open, and the framework is implemented in a freely available online calculator, so the analysis can be reproduced, re-parameterised and extended as new data arrive.

Section 2 relates the paper to the literature. Section 3 describes the data and the representative-dwelling construction. Section 4 sets out the simulation framework. Section 5 presents the historical results, Section 6 the required appreciation rate, and Section 7 the robustness analysis. Section 8 discusses interpretation and limitations, and Section 9 concludes.

2 Related literature

The paper draws on, and speaks to, five strands of work.

User cost and tenure choice. The theoretical benchmark for rent-versus-own comparisons is the user cost of owner-occupied housing — foregone interest on equity, mortgage interest, maintenance and depreciation, less expected capital gains — formalised by Poterba (1984) on tenure-choice foundations laid by Henderson and Ioannides (1983), and turned into the standard housing-valuation tool by Himmelberg, Mayer and Sinai (2005). A measurement branch shows, however, that ex-ante user costs diverge persistently from observed rents and are acutely sensitive to imputed appreciation and financing assumptions (Verbrugge, 2008; Díaz and Luengo-Prado, 2008), and that credible price-rent comparisons require quality- and composition-matched data (Hill and Syed, 2016; Bracke, 2015). This paper is the ex-post counterpart to that ex-ante tradition: rather than imputing an equilibrium user cost, I track the realised cash flows of both tenures month by month, which sidesteps the imputation problem; my required appreciation rate is the user-cost equilibrium condition solved for the capital-gains term rather than assuming it; and the composition-consistent national aggregate of Section 3.2 responds directly to the matched-measurement critique — the official England price and rent aggregates, I show, fail exactly that test.

Buy-versus-rent horse races. The closest methodological ancestors are the ex-post backtests for the United States: Beracha and Johnson (2012), who run cash-flow-matched buy-versus-rent races over 1978–2009 and find renting frequently competitive; the companion indifference-appreciation exercise of Beracha, Seiler and Johnson (2012), an antecedent of my RAR metric computed under stylised ex-ante assumptions rather than realised paths; and Rappaport (2010) and Goodman and Mayer (2018) on homeownership as wealth-building, with Herbert, McCue and Sanchez-Moyano (2013) surveying the distributional evidence. For Britain, the only comparable academic exercise is Mostafa and Jones (2019), an annual-frequency comparison of first-time-buyer cohorts over 1975–2011 in which the renter's counterfactual earns building-society deposit rates; buying wins essentially always. Tabner (2016) provides a UK net-present-value framework under assumed parameters, and practitioner analyses exist for Canada (Felix and Arif, 2025). Relative to this work, the present paper is — to my knowledge — the first monthly, cash-flow-matched backtest on UK data, and the first in which the British renter's counterfactual is a diversified equity portfolio; that single change overturns the always-buy conclusion and produces near-parity, while extending the sample through the post-2012 boom, the pandemic, and the 2022–2023 rate shock that earlier UK samples miss.

Housing returns. A recent measurement literature quantifies long-run returns to residential property: house prices track incomes rather than compounding like equities over the long run (Knoll, Schularick and Steger, 2017); total returns are competitive with equities only once rental yield is included and before costs (Jordà et al., 2019); and micro-level studies show that transaction costs, management and idiosyncratic risk substantially erode net returns to individual owners (Chambers, Spaenjers and Steiner, 2021; Eichholtz et al., 2021; Giacoletti, 2021; Demers and Eisfeldt, 2022), with large spatial heterogeneity (Amaral et al., 2025). My results are the household-level, England-specific counterpart: with realistic costs the unlevered housing return loses badly to equities — an all-cash buyer never wins — and it is mortgage leverage, not the housing return itself, that makes ownership financially competitive.

Housing in household portfolios. A structural literature examines how owner-occupied housing shapes household portfolios and lifecycle choices (Flavin and Yamashita, 2002; Cocco, 2005; Yao and Zhang, 2005; Chetty, Sandor and Szeidl, 2017; Vestman, 2019); J. Y. Campbell (2006) and Gomes, Haliassos and Ramadorai (2021) survey the field. These models emphasise precisely the two forces my backtest measures ex post: the portfolio distortion of a large, levered, indivisible housing position, and the financing channel through which mortgage debt amplifies returns to home equity. The empirical leverage gradient I document — buying wins two-thirds of cohorts at 95% LTV and none at all in cash — is, in effect, a realised-data trace of the mechanism these models study.

Rent risk, commitment, and the UK market. Two channels are deliberately switched off in my accounting and frame its interpretation. Ownership hedges rent risk (Sinai and Souleles, 2005), while my renter bears actual market rents throughout; and mortgage amortisation acts as forced saving for present-biased households (Laibson, 1997; Thaler and Benartzi, 2004; Schlafmann, 2021), with Bernstein and Koudijs (2024) showing amortisation causally builds net worth and Sodini et al. (2023) identifying substantial benefits of ownership quasi-experimentally — whereas my renter is endowed with perfect investment discipline. Near-parity for a perfectly disciplined renter therefore implies an advantage for ownership among typical households, which is how I read the result. Finally, the paper connects to UK housing economics — cycles and regional dynamics (Muellbauer and Murphy, 1997; Meen, 1999), rate-driven valuations (Miles and Monro, 2021), supply constraints (Hilber and Vermeulen, 2016), and the stamp-duty responses documented by Besley, Meads and Surico (2014) and Best and Kleven (2018), which motivate my period-accurate SDLT module — and to the evidence that price-rent ratios forecast housing returns (Case and Shiller, 1989; J. Y. Campbell and Shiller, 1988; Gallin, 2008; S. D. Campbell et al., 2009; Engsted and Pedersen, 2015): my finding that the entry price-to-rent ratio and mortgage rate explain 74% of cohort outcomes is the cash-flow-matched analogue of that predictability.

3 Data

The analysis draws on official statistics for all housing inputs and on index-provider data for financial returns. All series are monthly, in nominal pounds sterling, and cover January 2005 to April 2026 (256 months). I focus on England and its nine regions: England is the only part of the United Kingdom for which official monthly rent statistics reach back to 2005. Appendix A provides construction details; Table 1 summarises the sources.

Table 1: Data sources

Series	Source	Coverage used
House prices	UK House Price Index, HM Land Registry (all dwellings, England and regions)	2005:1–2026:4
Private rents	ONS Price Index of Private Rents (PIPR) and its historical series	2005:1–2026:4
Mortgage rate	Bank of England effective rate on new advances (CFMBJ95)	2005:1–2026:4
Equity returns	MSCI ACWI net total return, GBP, less 0.12% p.a. fund fee	2005:1–2026:4
Gilt yield	10-year UK government bond yield (OECD, via FRED)	2005:1–2026:4
Stamp duty	HMRC SDLT schedules, first-time-buyer rules by month	2003:12–2026:4

3.1 House prices and rents

House prices come from the UK House Price Index (UK HPI), the official transaction-based index constructed from HM Land Registry sales records by hedonic regression with geometric averaging (HM Land Registry, 2026). I use the average price of all dwellings, monthly, for England and each region.

Rents come from the ONS Price Index of Private Rents (PIPR), the official measure of private rents based on Valuation Office Agency lettings data (Office for National Statistics, 2026a; Office for National Statistics, 2026b). Two features make PIPR the appropriate counterpart to the UK HPI. First, it publishes average rent *levels* in pounds per calendar month, not only an index. Second, it is a *stock* measure covering new and existing tenancies, matching the position of a long-term renter whose rent adjusts with the market rather than only at new lets. The live PIPR series begins in January 2015; earlier years use the ONS historical series, chain-linked to PIPR's predecessor index back to January 2005 for England and its regions (Office for National Statistics, 2025b). The two sources agree exactly over their 122-month overlap, so the splice introduces no discontinuity.

3.2 A composition-consistent representative dwelling

Moving from regions to a single national figure raises a measurement issue that, to my knowledge, has not been documented. The headline UK HPI England price and the headline PIPR England rent each average the country, but they weight the regions differently: the price is a transaction-mix average, close to a simple average of the nine regions, whereas the rent is a tenancy-mix average dominated by London's large private rental sector. In April 2026 the national average rent (£1,438) exceeded the average rent of every region except London, while the national average price (£291,445) sat near the middle of the regional range. The ratio of the two — a price-to-rent ratio of 16.9 — consequently falls *below the ratio of every individual region* (17.5 to 22.2). Any rent-versus-own comparison built on this pair prices a mid-England home against a London-weighted rent, overstating the cost of renting and biasing the comparison towards ownership.

I therefore construct the price and rent of a *representative English dwelling* as the same population-weighted average of the nine regions:

$$P_t^{\text{rep}} = \sum_{i=1}^{9} w_i \, P_{i,t}, \qquad R_t^{\text{rep}} = \sum_{i=1}^{9} w_i \, R_{i,t}, \qquad w_i = \frac{\text{pop}_i}{\sum_j \text{pop}_j}, \tag{1}$$

with fixed weights from mid-2023 ONS regional population estimates. Both aggregates then share one regional composition, and the resulting price-to-rent ratio (20.7 in April 2026) lies within the regional range. Over the sample the representative dwelling appreciated from £162,312 to £317,738 (3.2% per year), falling 18% peak-to-trough during 2007–2009, while its rent rose from £692 to £1,280 per month (2.9% per year). The regional analyses in Sections 5 and 6 use each region's own price and rent and are unaffected by aggregation.

3.3 Mortgage rates

The mortgage rate is the Bank of England's effective interest rate on new mortgage advances to households (series CFMBJ95): the volume-weighted average rate actually paid on newly drawn mortgages each month, across products, fixation lengths and loan-to-value tiers (Bank of England, 2026). I prefer it to the Bank's quoted (advertised) series, which exist only for particular product tiers and have multi-year gaps around the financial crisis. Over the sample the effective rate averaged 3.5%, peaking at 6.1% in August 2008, reaching a low of 1.5% in November 2021, and standing at 4.1% in April 2026. Because first-time buyers borrow at higher loan-to-value ratios than the average borrower, they typically pay a premium over this series; Section 7 shows results with a 40 basis-point premium.

3.4 Financial returns

The renter's portfolio tracks the MSCI ACWI index, covering large- and mid-capitalisation equities in 47 developed and emerging markets (MSCI Inc., 2026). I use end-of-month index levels computed by MSCI directly in pounds sterling in the net-total-return variant (dividends reinvested after withholding taxes), less an ongoing charge of 0.12% per year representing a low-cost global index fund available to UK retail investors. Measuring the index in sterling embeds the currency exposure an unhedged UK investor bears. Over the sample the net-of-fee series returned 10.2% per year (a factor of 7.85). As a home-bias robustness check I also use a FTSE 100 total-return series, which returned 7.1% per year. The 10-year gilt yield (OECD, via FRED) serves as a conservative alternative opportunity cost in Section 6; it ranged from 5.4% (2007) to 0.2% (July 2020) and ended the sample at 4.8% (Federal Reserve Bank of St. Louis, 2026).

Figure 1 shows the three housing inputs; Figure 2 compares cumulative asset performance. The gap the buyer must overcome is evident: over twenty-one years the global equity portfolio multiplied roughly eightfold, UK large-cap equities fourfold, and the representative English dwelling — before the owner's running costs — twofold.

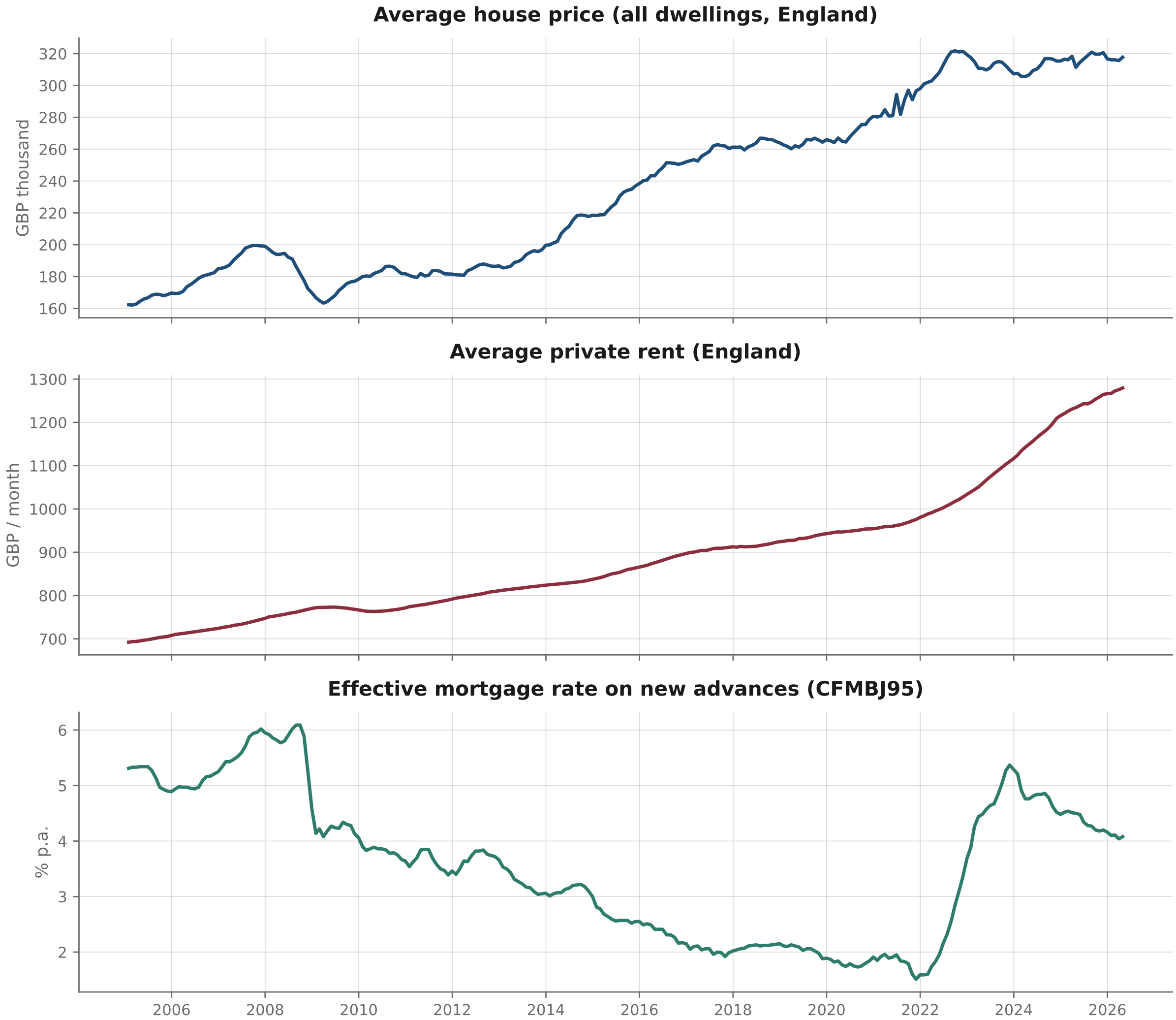

Figure 1: Housing-market inputs, representative English dwelling, 2005–2026. *Top:* average house price. *Middle:* average private rent. *Bottom:* Bank of England effective mortgage rate on new advances. Sources: HM Land Registry UK HPI; ONS PIPR; Bank of England.

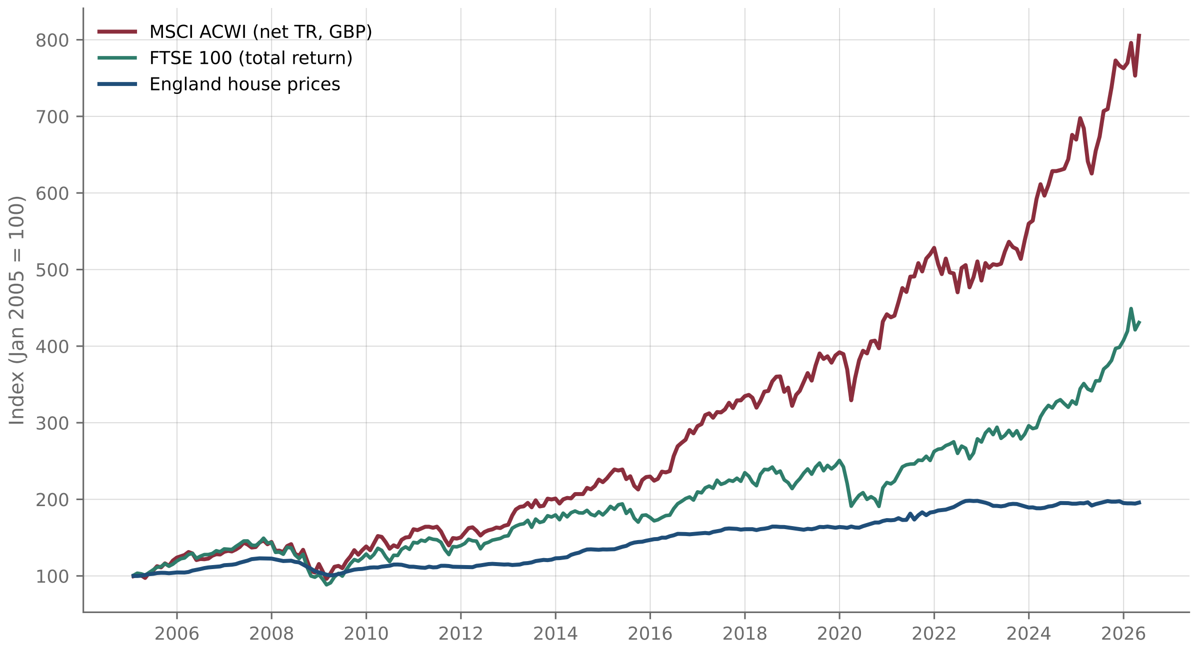

Figure 2: Cumulative performance of global equities (MSCI ACWI, net total return in GBP), UK equities (FTSE 100 total return) and representative English house prices, January 2005 = 100. Sources: MSCI; Morningstar; HM Land Registry.

4 Empirical framework

The simulation compares two housing strategies on a cash-flow-matched basis: both start with identical capital, and their cash flows are equalised every month. Whenever the renter pays less for housing in a given month, the difference is invested; whenever the renter pays more, the difference is withdrawn from the portfolio. The design follows Beracha and Johnson (2012), with transaction costs and taxes calibrated to the English first-time buyer.

4.1 The buyer

The buyer purchases the property at the market price prevailing in the entry month, with a deposit of 10% of the price (loan-to-value of 90%), purchase costs of 1.5% of the price (conveyancing, survey, mortgage and valuation fees), and stamp duty under the first-time-buyer schedule in force in that month. The mortgage amortises over 30 years. In the baseline the rate floats: each month the payment is recalculated at the prevailing effective market rate over the remaining balance and term,

$$r_{m,t} = (1 + r_t)^{1/12} - 1, \qquad \text{Payment}_t = \frac{B_t \, r_{m,t}}{1 - (1 + r_{m,t})^{-(T-t)}}, \tag{2}$$

where r_t is the annual effective rate, B_t the outstanding balance and $T - t$ the remaining months. Fixed-rate products of two, five and ten years are examined in Section 7.

Each month the owner also pays maintenance and depreciation of 1.5% per year of the property's current market value,

$$M_t = \frac{0.015}{12} V_t, \tag{3}$$

an all-in charge covering routine maintenance, major works and replacements, buildings insurance, and physical depreciation not offset by spending. The calibration sits between measured owner-occupier cash outlays on repairs and improvements — roughly 1.1–1.3% of the average house value per year in ONS Family Spending data (Office for National Statistics, 2025a) — and the 2–3% range the user-cost literature reaches once insurance and unrecovered depreciation are included (Miles and Monro, 2021; Judge and Leslie, 2021). Because this parameter proves the single most consequential assumption, Section 7 reports results for 1% through 3%. The property value itself evolves with the relevant house-price index, $V_{t+1} = V_t(1 + g_t)$.

The buyer's terminal wealth after a holding period ending at T^* is net housing equity after selling costs of 2% (estate-agent commission, conveyancing, energy certificate):

$$W^{\text{own}} = V_{T^*}(1 - 0.02) - B_{T^*}. \tag{4}$$

4.2 The renter

The renter occupies an equivalent dwelling at the market rent prevailing each month and invests all spare cash. The initial portfolio equals the buyer's total initial outlay — deposit plus purchase costs plus stamp duty — and each month the difference between the owner's outflow (mortgage payment plus maintenance) and the rent is added to, or withdrawn from, the portfolio, which earns the monthly return on the global equity index (net of withholding taxes and fund fees). The renter's terminal wealth is the portfolio value, $W^{\text{rent}} = P_{T^*}$. In no simulation does the renter's portfolio hit zero; the strategy is self-financing throughout.

4.3 Outcome measure and analyses

Outcomes are summarised by the renter/owner wealth ratio

$$R = \frac{W^{\text{rent}}}{W^{\text{own}}}, \tag{5}$$

following Felix and Arif (2025); $R > 1$ means renting-and-investing produced greater terminal wealth. The ratio is invariant to the initial capital scale and comparable across cohorts and regions.

Four analyses are run: (i) a single full-sample race from January 2005; (ii) 85 rolling quarterly entry cohorts, January 2005 to January 2026, all evaluated in April 2026; (iii) the same analyses region by region, aggregated with population weights; and (iv) the required appreciation rate of Section 6, computed ex post over fixed ten-year holds and ex ante at end-of-sample conditions.

4.4 Stamp duty

Stamp Duty Land Tax is modelled period-accurately: each simulated purchase pays the tax implied by the England first-time-buyer schedule in force in that month — including the 2008–2009 holiday, the 2010–2012 first-time-buyer relief, the 2014 slab-to-slice reform, the permanent relief from November 2017, the 2020–2021 holiday, and the 2022–2025 threshold changes (HM Revenue & Customs, 2026; House of Commons Library, 2025). Appendix B tabulates the schedules. For the representative dwelling, a first-time buyer paid SDLT in 141 of 256 purchase months (55%), averaging roughly £1,800 (0.9% of the price) when payable.

4.5 Scope and abstractions

The model excludes several factors by design. Income and capital-gains taxes are omitted on both sides: a first-time buyer's main residence is exempt from capital-gains tax, and the renter's flows fit within annual ISA allowances for most of the sample, so the symmetric no-tax treatment is a reasonable approximation for the typical household (though not for very large portfolios held outside tax wrappers). Council tax and utilities are paid by the occupant under both tenures and net out. The renter never borrows to invest, so leverage is available only through the mortgage. Non-financial considerations — security of tenure, freedom to modify, flexibility, the insurance value of ownership against rent risk (Sinai and Souleles, 2005) — are outside the accounting and are discussed in Section 8. Table 2 collects the baseline parameters.

Table 2: Baseline parameters (first-time buyer)

Parameter	Description	Value
Property	Representative English dwelling (Section 3.2)	—
Deposit	Share of purchase price	10%
Mortgage term	Amortisation period	30 years
Rate type	Variable, recast monthly at CFMBJ95	—
Purchase costs	Conveyancing, survey, mortgage fees	1.5%
Stamp duty	England first-time-buyer schedule, by month	rule-based
Selling costs	Estate agent, conveyancing, EPC	2%
Maintenance & depreciation	Share of current market value, per year	1.5%
Equity index	MSCI ACWI net total return, GBP, less 0.12% fee	—
Rent	Market rent, updated monthly	—
Currency	Nominal GBP throughout	—

5 Results

5.1 The full-period race

I first run the longest race the data allow: a buyer and a renter who both start in January 2005 and are evaluated in April 2026. The buyer purchases the representative dwelling for £162,312 with a deposit of £16,231, purchase costs of £2,435 and stamp duty of £1,623; the renter starts with the identical £20,289 invested in the equity index and occupies the same dwelling at the market rent of £692 per month. Table 3 reports the outcome and Figure 3 the two wealth paths.

Table 3: Full-period simulation, January 2005 – April 2026

	GBP
Purchase price, January 2005	162,312
Initial capital (deposit + costs + SDLT)	20,289
Property value, April 2026	317,738
Remaining mortgage balance	58,615
Selling costs (2%)	6,355
Owner's terminal wealth	252,768
Renter's terminal wealth	275,171
Difference (renter − owner)	22,403
Wealth ratio R	1.09

Notes: Baseline parameters of Table 2. The renter's portfolio remains positive throughout; no external cash is required by either strategy beyond the matched flows.

After twenty-one years the renter finishes 9% ahead ($R = 1.09$) — a close race whose lead changes hands more than once. The owner, holding a leveraged position in a rising asset, led through much of the 2010s before the renter's portfolio pulled narrowly ahead in the equity-strong years after 2020. Figure 4 shows the matched cash flows: the owner's outlay exceeded the rent in 173 of 256 months, so the renter added to the portfolio in most months, but the margins are modest — the mortgage payment starts at £801 against rent of £692, and maintenance grows from £203 to £397 per month as the property appreciates. That a leveraged claim on an asset appreciating at 3.2% per year very nearly matched a portfolio compounding at 10.2% quantifies how much work the mortgage does for the buyer.

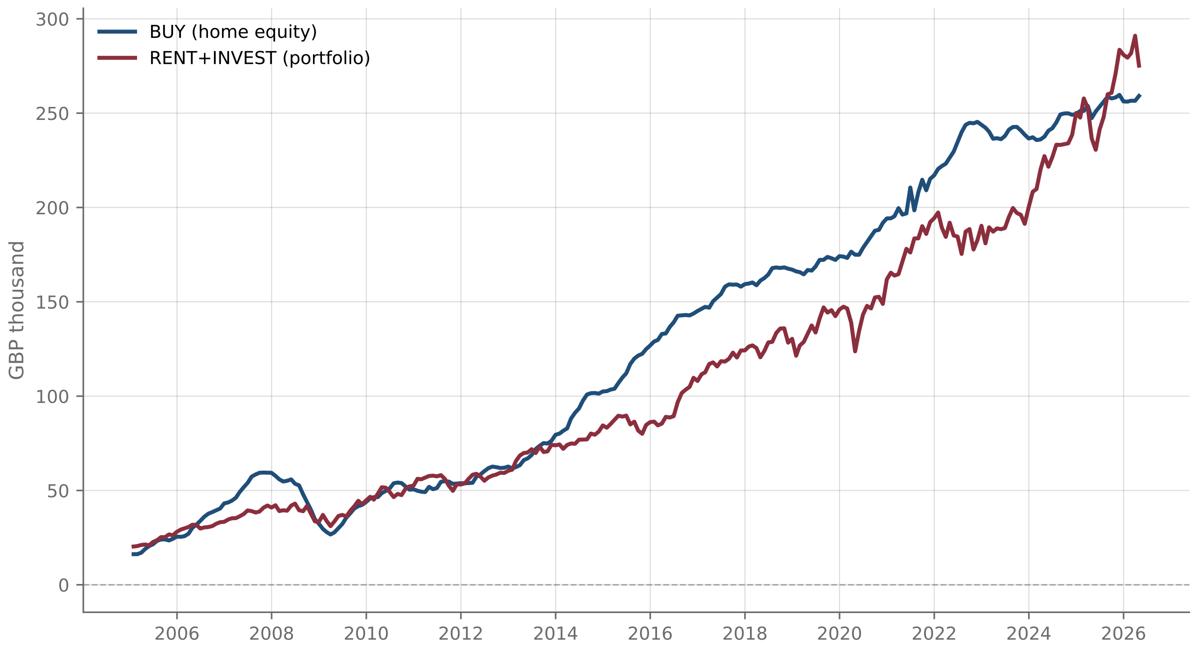

Figure 3: Owner's home equity and renter's portfolio value, January 2005 entry cohort.

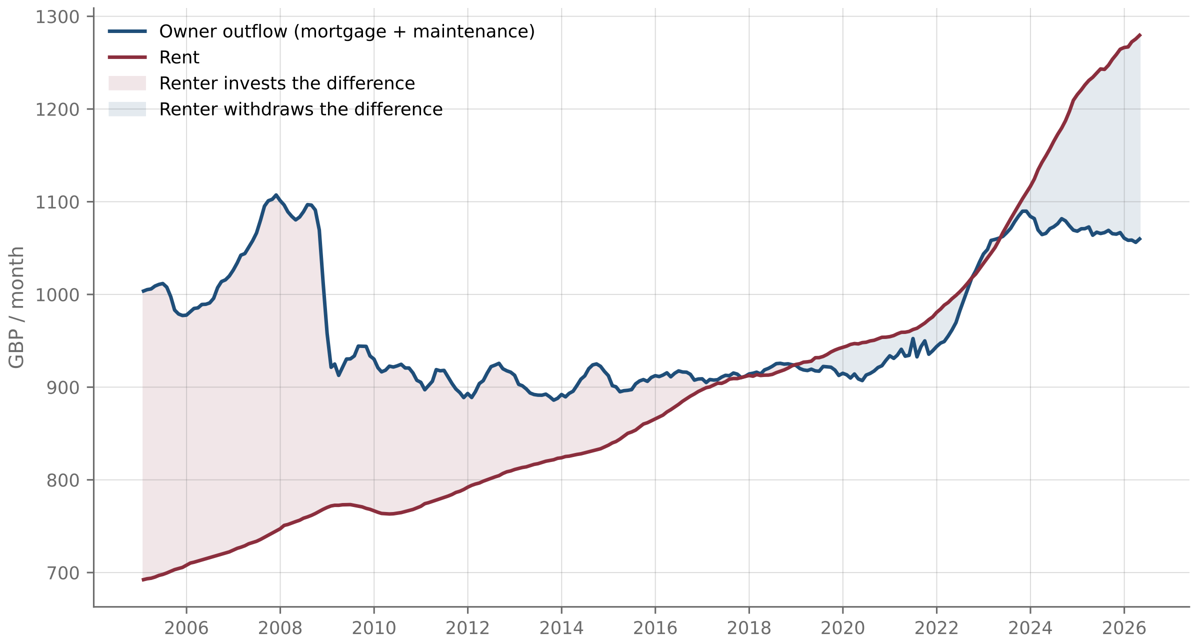

Figure 4: Monthly housing outlays of the owner (mortgage payment plus maintenance) and the renter (market rent), January 2005 entry cohort. Shading indicates months in which the renter invests (owner pays more) or withdraws (renter pays more).

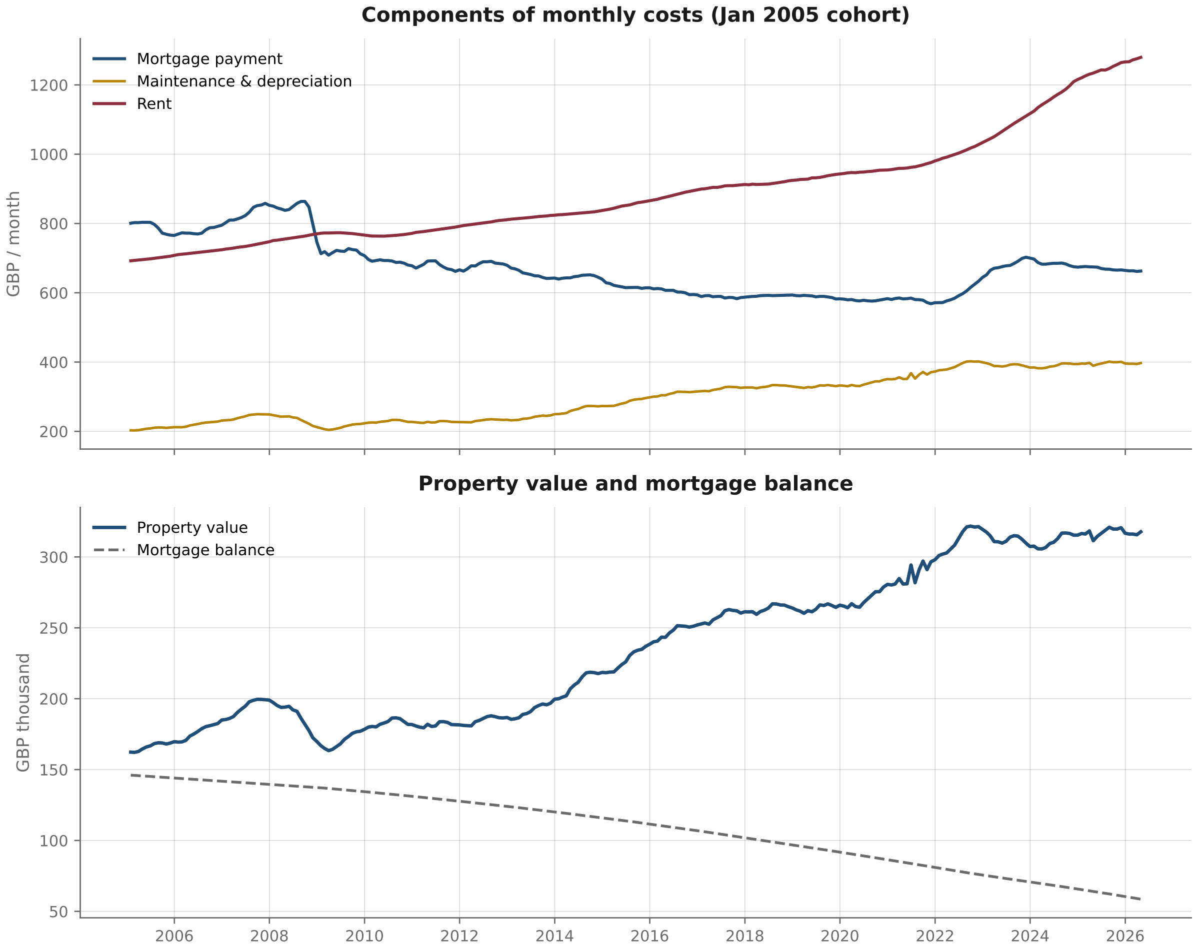

Figure 5: Cost components and balance-sheet evolution, January 2005 entry cohort. The variable-rate payment declines with rates and amortisation while maintenance and rent rise; with a 30-year term the mortgage is only partly repaid by April 2026.

5.2 Rolling entry cohorts

The single 2005 start is one draw from history. To isolate timing, I repeat the race for 85 cohorts entering in every quarter from January 2005 to January 2026, all evaluated in April 2026. Figure 6 plots the wealth ratio by entry date.

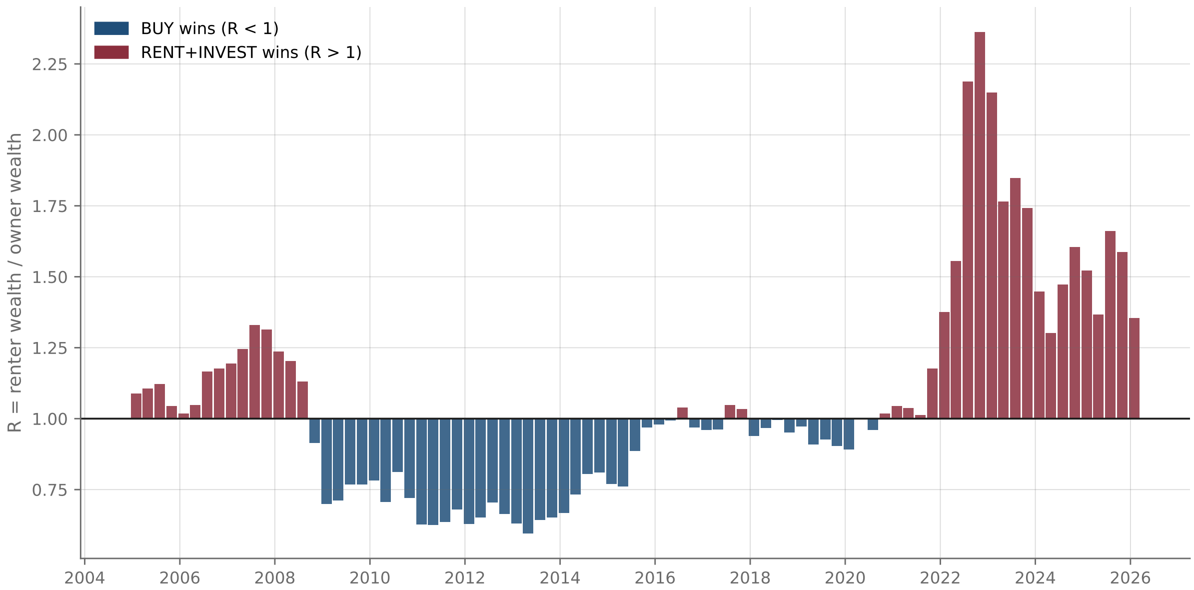

Figure 6: Renter/owner wealth ratio R for 85 quarterly entry cohorts, all evaluated in April 2026. Values above one indicate that renting-and-investing produced greater terminal wealth.

Buying produced greater wealth in 45 of the 85 cohorts (53%) and renting in 40 (47%); the mean wealth ratio is $\bar{R} = 1.06$ and the median 0.99 — collectively, a dead heat. Around the parity, however, dispersion is wide and highly systematic. The best cohort for buying (April 2013) left the renter with 60% of the owner's wealth; the best for renting (October 2022) left the renter 2.4 times as wealthy. The cohorts sort into three regimes: entries during 2005–mid-2008, at pre-crisis prices, lost to the renter; almost every entry from late 2008 through 2020 — post-crisis prices financed by a decade of cheap and falling-rate debt — beat the renter; and every entry since late 2020 has so far lost, as the equity surge and the 2022–2023 rate shock reversed the conditions. Stamp duty plays a minor role throughout: 48 of 85 cohorts paid any SDLT, never exceeding roughly 1% of the purchase price.

5.3 Regional results

Running the race region by region — each region with its own price *and* rent, so each internally consistent — both validates the representative-dwelling aggregate and reveals the geography of the outcome. Table 4 and Figure 7 report the single-start wealth ratio, the share of cohorts won by buying, and the mean historical required appreciation rate (defined in Section 6) for each region.

Table 4: Regional outcomes, 2005–2026

Region	Price (2026:4)	P/R	R (2005)	Buy-win %	Hist. RAR
North East	£163,190	17.5	1.36	67	1.2%
North West	£216,138	18.9	0.85	73	0.8%
Yorkshire and The Humber	£207,974	20.3	1.03	67	1.2%
West Midlands	£234,635	20.3	1.10	65	1.0%
South West	£302,618	20.5	1.48	59	2.3%
London	£552,655	20.1	0.63	39	1.7%
East Midlands	£241,620	22.1	1.48	60	2.3%
East of England	£336,300	21.9	1.32	33	2.3%
South East	£376,819	22.2	1.37	26	2.6%
England (pop.-weighted)	—	20.6	1.14	51	1.7%

Notes: R (2005) is the wealth ratio of the January 2005 entry cohort ($R > 1$: renting won). Buy-win % is the share of 85 quarterly cohorts in which buying produced greater wealth. Hist. RAR is the mean, across ten-year holds, of the annual appreciation required for buying to break even against the equity portfolio (Section 6). P/R is the April 2026 ratio of price to annual rent.

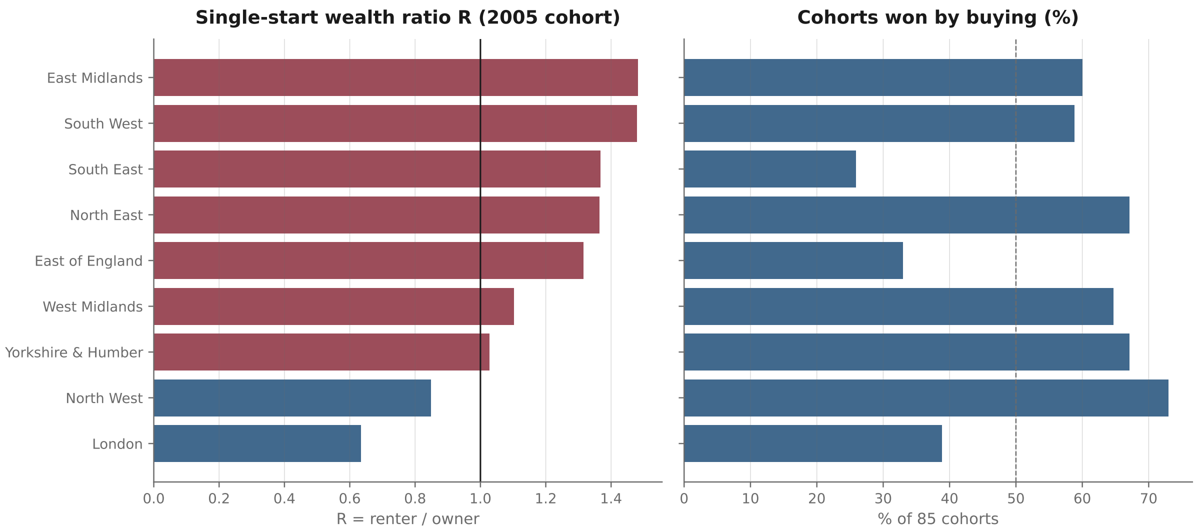

Figure 7: Regional outcomes: single-start wealth ratio, January 2005 cohort (left) and share of the 85 entry cohorts won by buying (right).

The population-weighted aggregates (R of 1.14; buy-win share of 51%) bracket the representative-dwelling figures (1.09; 53%), confirming that the aggregation is innocuous once composition is held consistent. The cross-sectional pattern tracks affordability rather than raw price growth: in the North and Midlands, where price-to-rent ratios are lowest, owning is cheap relative to renting and buying won roughly two-thirds of cohorts; in the South and East, where the ratios exceed 22, buying won a quarter to a third. London is distinctive: a 10% deposit on a £553,000 dwelling is a large leveraged position, and strong price growth made buying the clear winner of the 2005 start ($R = 0.63$) even though it won only 39% of cohorts overall.

5.4 What predicts the outcome

Which conditions observable at purchase predict the eventual winner? Regressing the cohort wealth ratio on the entry-month mortgage rate and price-to-rent ratio yields

	Pearson r	R^2	p
Mortgage rate	0.22	0.05	0.045
Price-to-rent ratio	0.59	0.35	<0.001
Joint (OLS)	—	0.74	<0.001

Neither variable predicts strongly alone, but jointly they explain 74% of the cross-cohort variation: valuation and financing cost together summarise most of what a buyer needed to know at the moment of decision. Two threshold regularities follow (Figures 8 and 9): buying won in every cohort that entered at a price-to-rent ratio below 19.5, and renting won in every cohort above 24.3. The representative ratio averaged 21.9 over the sample, peaked at 26.4 in August 2022, and stood at 20.7 in April 2026 — in the contested middle. Figure 10 shows the underlying dynamics; the mean reversion of the ratio around 22, and the dominance of its movements over those of the mortgage rate, are consistent with the evidence that rent-price ratios forecast housing returns (S. D. Campbell et al., 2009).

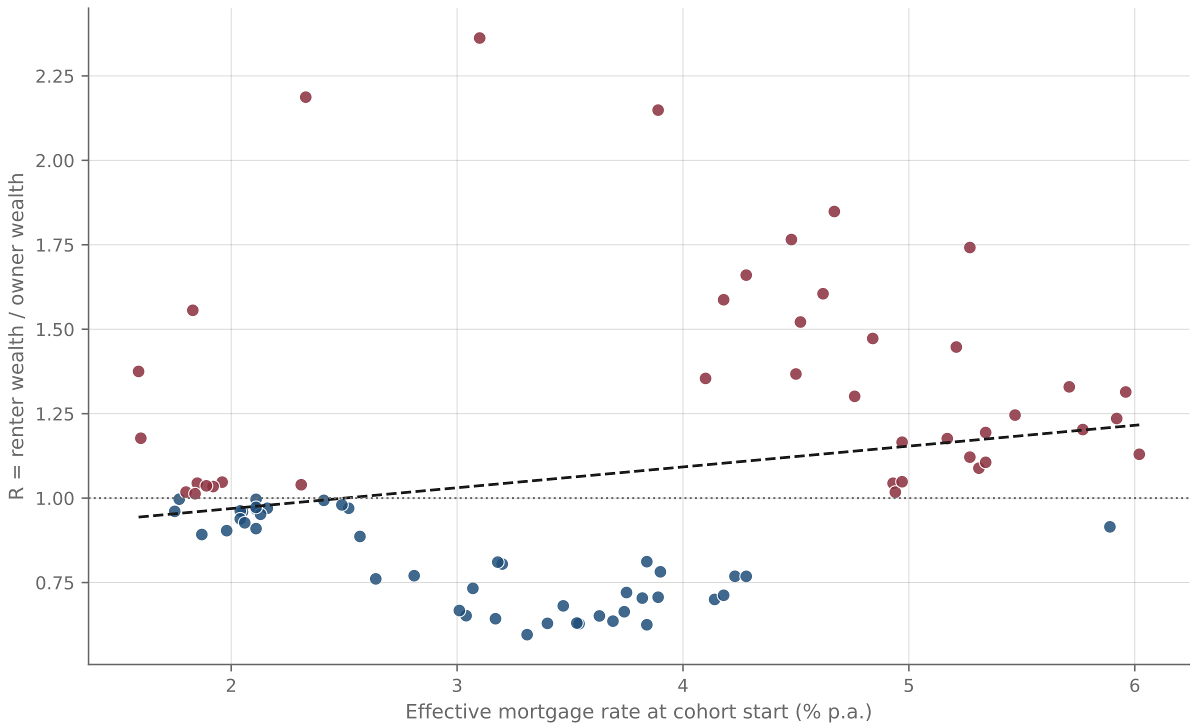

Figure 8: Effective mortgage rate at entry versus final wealth ratio, 85 cohorts.

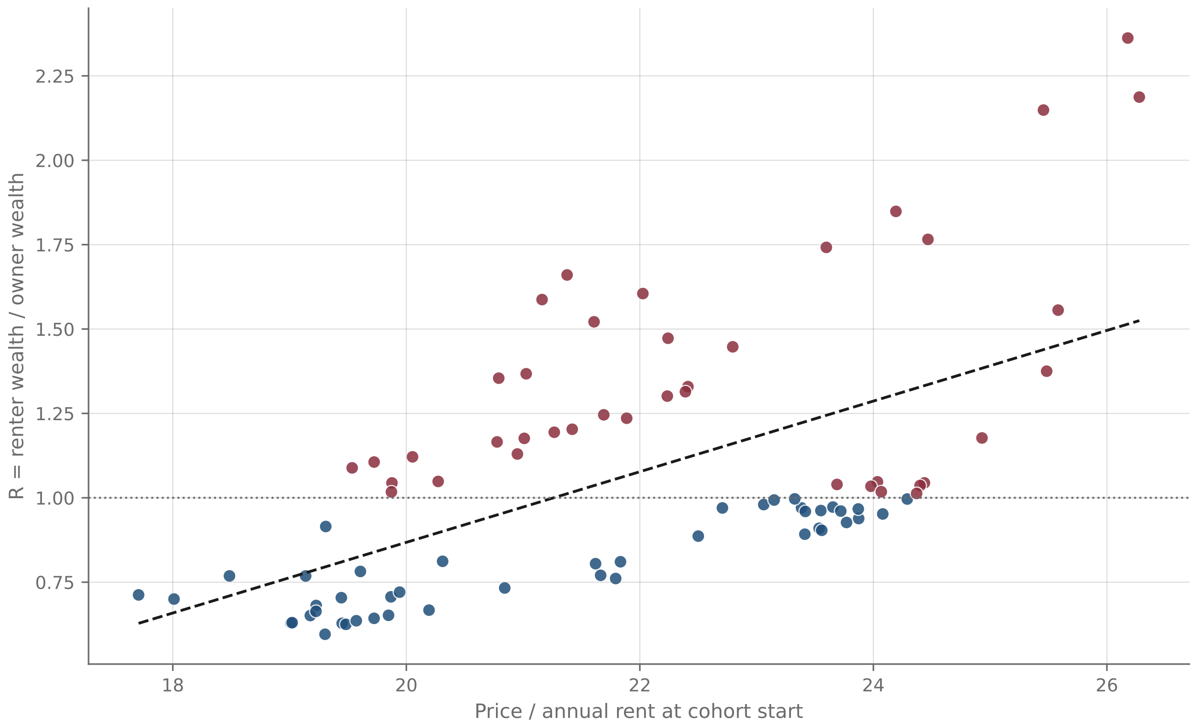

Figure 9: Price-to-rent ratio at entry versus final wealth ratio, 85 cohorts.

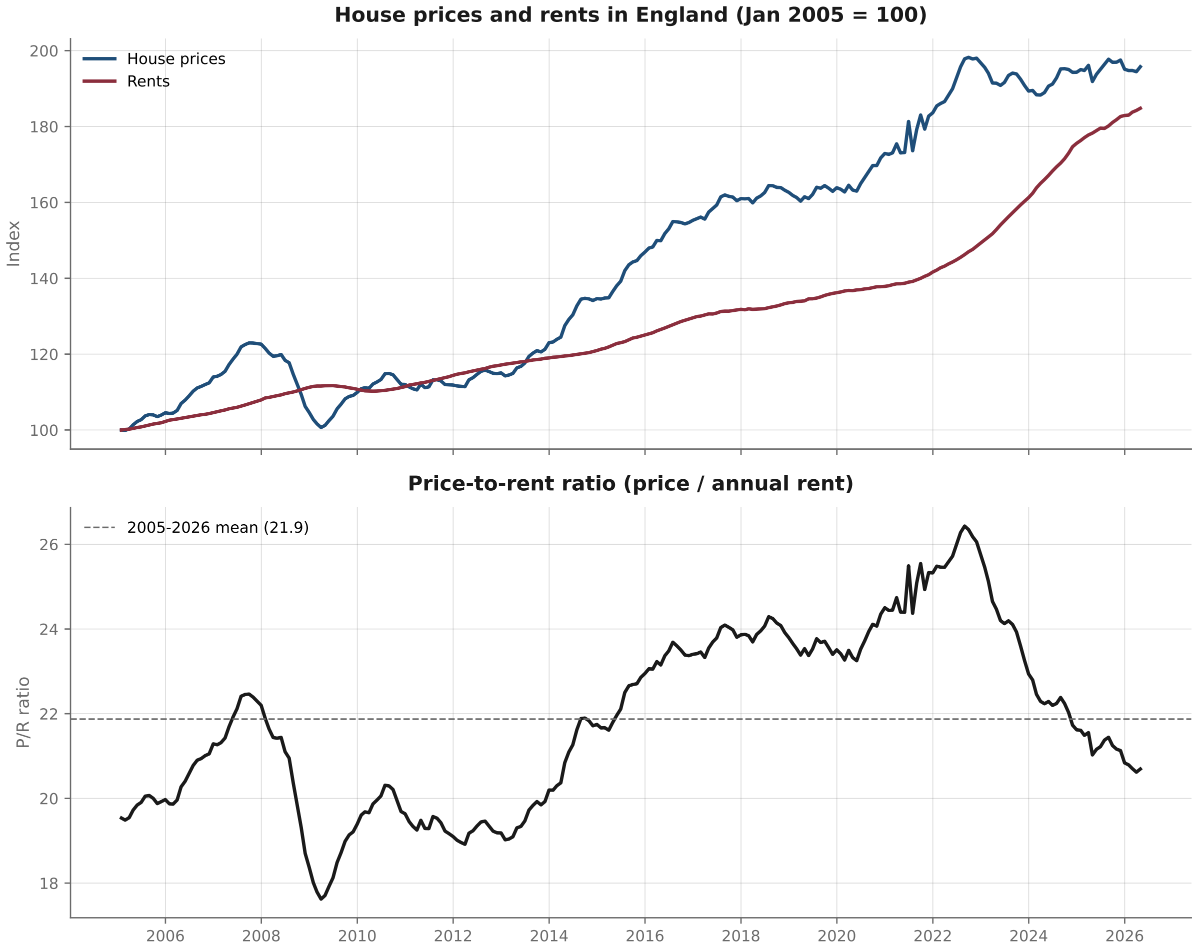

Figure 10: House prices and rents (January 2005 = 100) and the price-to-rent ratio, representative English dwelling.

6 The required appreciation rate

The cohort analysis looks backwards. To connect it to the decision facing a buyer at any point in time, I define the *required appreciation rate* (RAR): the constant annual house-price growth at which buying exactly matches renting-and-investing over a given horizon, holding all other inputs at their realised (ex post) or assumed (ex ante) paths. The RAR inverts the user-cost equilibrium of Poterba (1984) and Himmelberg, Mayer and Sinai (2005): rather than assuming an expected capital gain and asking whether prices are fair, it solves for the break-even capital gain and lets the reader supply their own expectation. If prices subsequently grow faster than the RAR, the buyer wins; slower, and the renter wins.

6.1 Historical RAR

For each entry month from January 2005 to April 2016 I compute the RAR over a fixed ten-year hold, using the realised paths of mortgage rates, rents and equity returns, and compare it with the appreciation actually delivered. Figure 11 plots both.

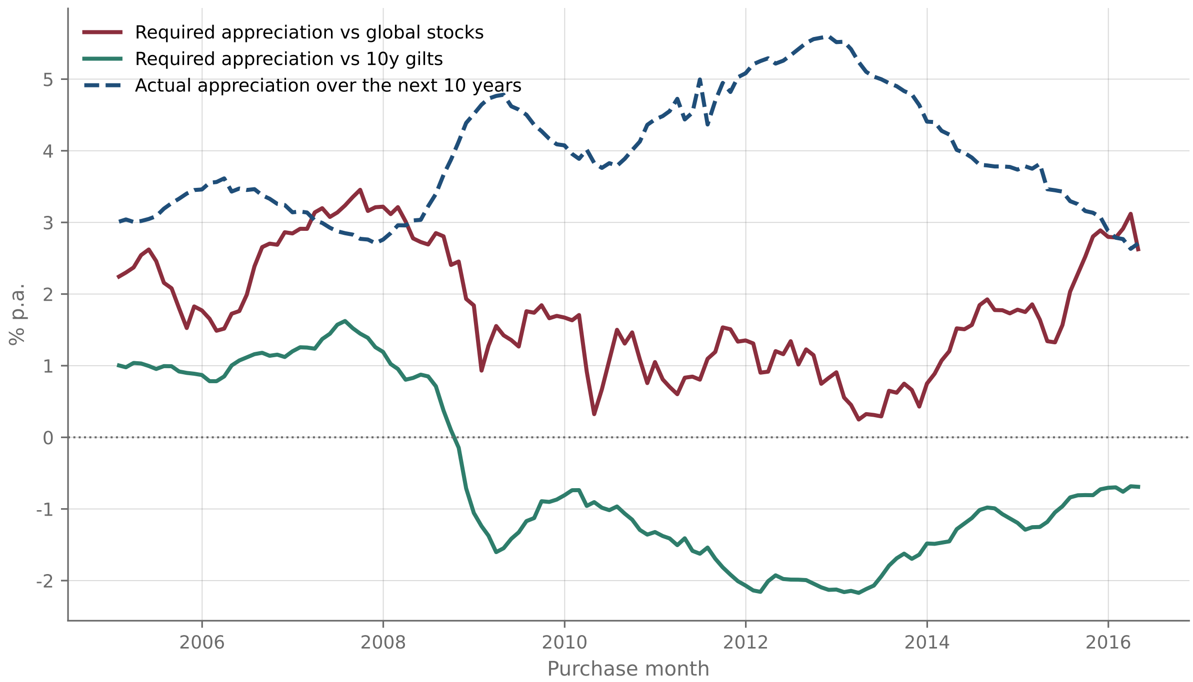

Figure 11: Required versus realised ten-year house-price appreciation, by entry month. The required rate is computed against the realised paths of mortgage rates, rents and either global equity returns or the ten-year gilt yield at entry.

Measured against the equity portfolio, the mean RAR across the 136 entry months is 1.8% per year (never above 3.5%), while the representative dwelling delivered 4.0% on average; buying was therefore, ex post, the better ten-year call in 88% of entry months. Measured against gilts the mean RAR is slightly negative (-0.6%): house prices could have drifted down and buying would still have beaten a bond-earning renter in most months. The apparent tension with the near-even cohort split of Section 5.2 is informative rather than contradictory: the rolling cohorts are all evaluated in April 2026 and are therefore weighted towards the recent equity-strong regime, whereas fixed ten-year holds end throughout 2015–2026. Horizon and end-date, not methodology, separate the two views.

6.2 Forward-looking RAR

At end-of-sample conditions — price £317,738, rent £1,280 per month (price-to-rent 20.7), effective mortgage rate 4.1%, first-time-buyer stamp duty of £887 — I compute the RAR under an expected equity return of 6.0% nominal less the 0.12% fee (5.9% net). The figure is deliberately conservative relative to the realised 10.2%: it combines a CAPE-implied long-run real return on global equities of roughly 3% — a real return on the same global portfolio, hence currency-invariant — with expected inflation of roughly 3%, the same assumption underlying the 3% rent-growth path. The current 4.8% gilt yield provides the conservative alternative.

Table 5: Forward-looking required appreciation rate, April 2026 conditions

Horizon	vs. equities (5.9%)	vs. gilts (4.8%)
3 years	2.2%	2.0%
5 years	1.6%	1.4%
7 years	1.3%	1.0%
10 years	1.0%	0.6%

Notes: Constant annual house-price growth at which buying matches renting-and-investing, at April 2026 inputs with 3% annual rent growth. Baseline parameters of Table 2.

Over a ten-year hold, the representative dwelling needs to appreciate only 1.0% per year to match the equity investor — well below the 3.2% it averaged over the sample and below most medium-term forecasts. The intuition is a user-cost identity. Renting costs the gross rental yield, 4.8% of the property's value per year at current prices. Owning costs mortgage interest on the 90% loan (about $4.1\% \times 0.9$), the opportunity cost of the 10% deposit, and 1.5% maintenance — together roughly 5.8% before appreciation. The 1-point gap is the appreciation buying needs; horizon effects (transaction-cost amortisation, mortgage paydown, rent growth against a nominally fixed payment schedule) account for the decline in the requirement as the hold lengthens.

6.3 Regional break-evens

Figure 12 maps the ten-year forward RAR against equities for the nine regions, using each region's April 2026 price and rent, the national mortgage rate, and region-specific stamp duty. The requirement ranges from -0.6% in the North East — where prices could fall and buying would still break even — to 1.6% in the South East. In every region the break-even lies below the region's own 2005–2026 average appreciation.

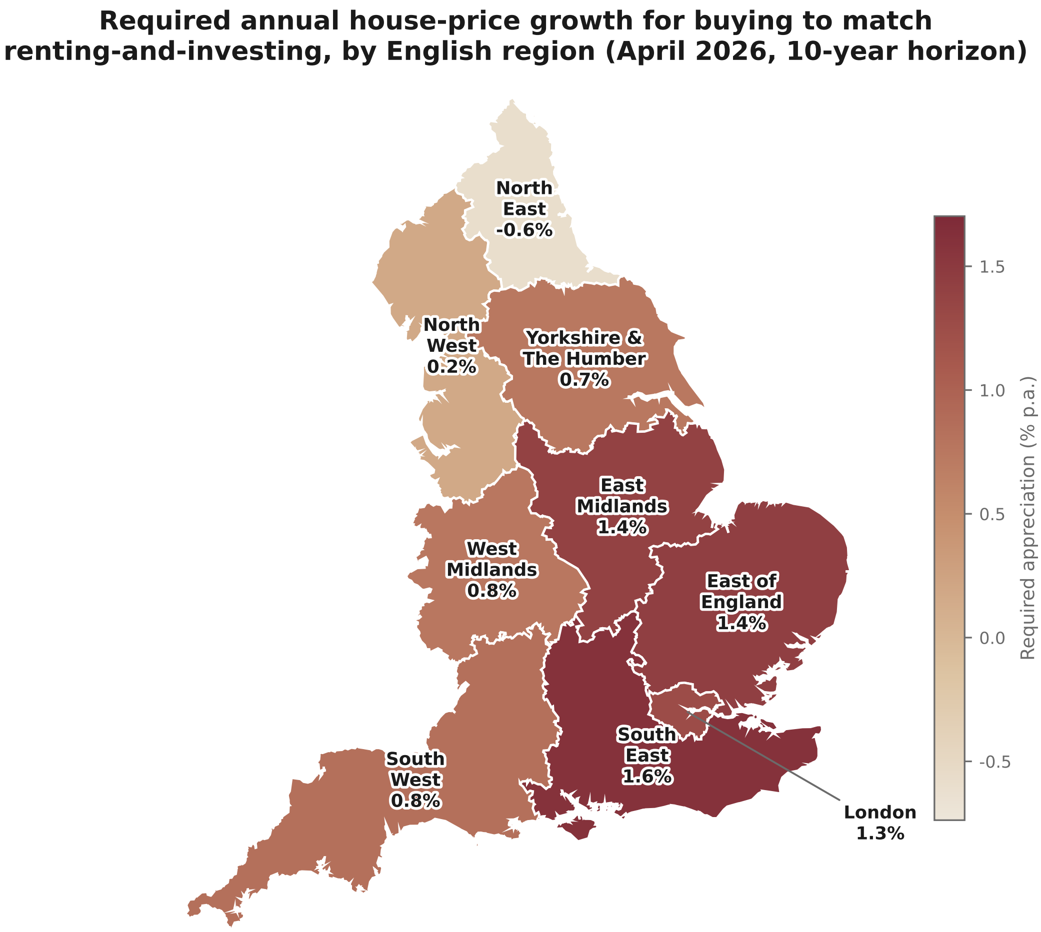

Figure 12: Forward-looking ten-year required appreciation rate against global equities, by region, April 2026 conditions.

7 Robustness

Tables 6 and 7 vary each baseline parameter in turn, reporting the full-period wealth ratio (January 2005 entry), the rolling-cohort mean, and the share of the 85 cohorts won by buying.

Table 6: Sensitivity to maintenance and deposit

Parameter	**Value**	R **(2005)**	$\bar{R}$	**Buy-win %**
Maintenance and depreciation (% of value p.a.)				
	1.0	0.78	0.88	76
	1.5 (baseline)	1.09	1.06	53
	2.0	1.39	1.24	26
	2.5	1.70	1.42	2
	3.0	2.01	1.60	0
Deposit (% of purchase price)				
	5	1.00	1.05	65
	10 (baseline)	1.09	1.06	53
	15	1.17	1.11	34
	20	1.25	1.16	25
	25	1.33	1.22	14
	50	1.70	1.45	0
	100 (cash)	2.32	1.79	0

Notes: $R > 1$: renting-and-investing produced greater wealth. $\bar{R}$ is the mean wealth ratio across the 85 quarterly cohorts; buy-win % the share won by buying. All other parameters at baseline.

Table 7: Sensitivity to other parameters

Parameter	Value	R (2005)	$\bar{R}$	Buy-win %
Mortgage term				
	25 years	1.22	1.12	40
	30 years (baseline)	1.09	1.06	53
	35 years	0.98	1.02	61
Rate fixation				
	Variable (baseline)	1.09	1.06	53
	2-year fixes	1.09	1.05	59
	5-year fixes	1.15	1.05	62
	10-year fixes	1.30	1.09	62
	+40bp high-LTV premium	1.21	1.15	35
Transaction costs				
	Purchase costs 1.0%	1.07	1.03	56
	Purchase costs 2.0%	1.11	1.09	48
	Selling costs 1.0%	1.08	1.02	56
	Selling costs 4.0%	1.12	1.16	39
	SDLT excluded	1.04	1.04	58
Renter's portfolio				
	MSCI ACWI net, −fee (baseline)	1.09	1.06	53
	MSCI ACWI gross, no fee	1.23	1.12	40
	FTSE 100 total return	0.62	0.91	73

Notes: As Table 6.

Four regularities organise the tables.

First, **maintenance is the single most consequential parameter**. Between 1% and 3% of value per year, the full-period ratio moves from 0.78 to 2.01 and the buy-win share from 76% to zero; within the defensible range the verdict shifts from "buying usually wins" (1%) through near-parity at the 1.5% baseline to "renting ahead" at 2%. The paper's central finding of near-parity is therefore conditional on this calibration, which I place between measured owner-occupier cash spending (≈1.2% of value) and the 2%-plus figures of the user-cost literature (Office for National Statistics, 2025a; Miles and Monro, 2021; Judge and Leslie, 2021). Readers with stronger priors can reposition the headline accordingly.

Second, **leverage is what keeps buying competitive**. A smaller deposit helps the buyer — at 5% down, buying wins 65% of cohorts; at 20%, 25%; and an unlevered cash buyer never wins ($R = 2.32$). Longer terms, which slow amortisation and prolong the leverage, work in the same direction. The first-time buyer's high loan-to-value is precisely what allows a claim on an asset appreciating at 3.2% to compete with a portfolio returning 10.2%; the result echoes the portfolio-choice observation that owner-occupied housing is attractive largely through its financing (Flavin and Yamashita, 2002).

Third, **transaction costs, stamp duty and fixation are second-order**: none moves the buy-win share by more than a handful of percentage points. England's low round-trip transaction costs are a structural advantage of ownership, and the period-accurate stamp-duty module, while conceptually important, changes little at average first-time-buyer prices. A 40 basis-point high-LTV rate premium costs the buyer visibly (buy-win 35%) but does not overturn the picture.

Fourth, **the renter's diversification is decisive**. Substituting the FTSE 100 for the global portfolio reverses the verdict ($R = 0.62$; buying wins 73% of cohorts): the three-point annual gap between UK large-cap and global equity returns compounds, over two decades, into the difference between the renter winning narrowly and losing badly. The renter's side of the parity is earned by disciplined, globally diversified investing, not by renting per se.

8 Discussion

8.1 International context

The near-parity result is a product of the particular history England delivered. Table 8 places English house-price growth in international context using BIS residential property price statistics (Bank for International Settlements, 2026). The United Kingdom sits mid-table over 2005–2025: cumulative nominal growth of 95% (3.3% per year), far below the central European booms and well below the pace that would have made ownership dominant. That a leveraged position in an asset growing 3.2% per year, carrying 1.5% running costs, kept pace with a global equity portfolio returning 10.2% is a measure of what mortgage leverage does for the household balance sheet; in a period kinder to housing or harsher to equities, buying would have won comfortably, and in the opposite case renting would have.

Table 8: Nominal house-price growth, selected countries, 2005Q1–2025Q4

Country	Cumulative	Annualised
Hungary	356%	7.6%
Portugal	156%	4.6%
Netherlands	110%	3.6%
United Kingdom	95%	3.3%
United States	89%	3.1%
Germany	85%	3.0%
Ireland	62%	2.3%
France	60%	2.3%
Spain	60%	2.3%
Italy	15%	0.7%

Source: BIS residential property prices, nominal, selected series. The UK entry covers the whole United Kingdom; the representative English series of this paper grew 96% (3.2% per year) over the same span.

8.2 Limitations

Several limitations qualify the results. *Aggregation.* The representative dwelling is a construct; households decide in a specific market, and the regional results of Section 5.3 — which vary from 26% to 73% buy-win shares — are the more decision-relevant numbers. Even within regions, composition differences between the owned and rented stocks mean the price-to-rent *level* carries measurement error that the *time variation* largely nets out. *Behavioural assumptions.* The renter invests 100% of every monthly difference, immediately and for two decades, in a globally diversified portfolio. Evidence on household saving and portfolio behaviour suggests this is an upper bound on typical conduct; the FTSE-only robustness check shows how costly even a diversification failure is, before any failure of discipline. *Taxes.* The symmetric no-tax treatment is appropriate for a typical first-time buyer (no capital-gains tax on a main residence; ISA capacity covering the renter's flows for most of the sample) but understates the drag on very large taxable portfolios, which would shade outcomes towards buying. *Inference.* The 85 cohorts overlap and share the sample's single realisation of asset returns; win shares and correlations describe this history and are not tests on independent draws. *Unpriced attributes.* Ownership insures against rent risk (Sinai and Souleles, 2005) and carries control rights; renting carries flexibility and mobility (Ministry of Housing, Communities and Local Government, 2025). None of these enters the terminal-wealth accounting.

8.3 Implications

Three implications follow for households and for the framing of housing-policy debates. First, the popular certainties on both sides are unwarranted: over a twenty-one-year window containing a housing crash, a decade of cheap credit and an exceptional equity bull market, buying and disciplined renting-and-investing produced nearly identical average wealth. Second, the decision-relevant information is observable at purchase: the price-to-rent ratio and the mortgage rate jointly explained 74% of the cross-cohort variation, and simple thresholds on the ratio separated the always-buy from the always-rent regions of the sample. Third, the comparison is conditional on conduct: the renter's side of the parity requires actually investing the cost difference, globally. A mortgage functions as a commitment device — amortisation is forced saving — and for households unlikely to match that discipline voluntarily, ownership is likely the safer default even where the pure arithmetic is a tie. This behavioural asymmetry, not the arithmetic, may be the strongest practical argument for buying.

9 Conclusion

This paper provides the first monthly, cash-flow-matched comparison of buying and renting-with-investing on English data, run for the first-time buyer over every quarterly entry point from January 2005 to January 2026, on a representative dwelling whose price and rent share one regional composition. The answer to the popular question is that, over these two decades, there was close to no answer: buying produced greater terminal wealth in 53% of entry cohorts, the mean renter/owner wealth ratio was 1.06, and the longest race ended with the disciplined renter 9% ahead. Entry conditions — summarised almost entirely by the price-to-rent ratio and the mortgage rate at purchase — determined which side of parity a given household landed on, with buying favoured after the 2008 crash and in the affordable North and Midlands, and renting favoured at the 2005–2008 peak, in the expensive South and East, and since 2020.

Beyond the headline, the analysis yields a methodological caution and a forward-looking metric. The caution is that the official national house-price and rent aggregates weight regions differently, so their ratio — the most widely quoted price-to-rent figure — lies outside the range of every English region and flatters ownership in any cost comparison built on it. The metric is the required appreciation rate: at April 2026 conditions, average English prices need grow only about 1% per year over a decade for buying to break even against a conservatively parameterised global equity alternative — below the sample-average appreciation, and negative in the North East. Whatever the last two decades favoured, the forward-looking arithmetic is close to balanced.

The results also discipline how the trade-off should be described. Leverage, not housing appreciation, is what keeps buying competitive: an unlevered buyer never beats the renter, and smaller deposits and longer terms systematically favour ownership. Conversely, the renter's parity is earned by conduct: it requires investing the full cost difference in a globally diversified portfolio, and substituting the home-market index alone reverses the verdict. The single most consequential calibration is the maintenance-and-depreciation charge, and I report the full mapping from that assumption to the headline so that readers can relocate the result under their own priors.

All data, code, and an interactive implementation of the framework are openly available, and the analysis is designed to be re-run as new months of official data arrive. Extensions to Scotland and Wales as their rent statistics mature, explicit tax modelling for high-saving renters, and first-time-buyer-specific price series are natural next steps.

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Appendices

A Data construction

Rents

The rent series splices two ONS products. The PIPR historical series (released March 2025) provides average monthly rents in GBP for England and the nine English regions from January 2005 to February 2025, chain-linked to the predecessor Index of Private Housing Rental Prices; the live PIPR monthly dataset provides the same measures from January 2015 onwards. Over their 122-month overlap (January 2015–February 2025) the two agree exactly at every observation, so the splice — historical values before January 2015, live values thereafter — introduces no discontinuity. Rent levels are model-consistent estimates published by the ONS (base-period mean rents uprated by the index), not raw sample means.

House prices

Regional and national average prices (all dwellings) are taken directly from the UK HPI full file (April 2026 vintage). The UK HPI revises its most recent months; the April 2026 observation used here is the first published estimate.

Representative dwelling

The representative price and rent are fixed-weight averages of the nine regional series with mid-2023 ONS population weights (millions): North East 2.65, North West 7.52, Yorkshire and The Humber 5.56, East Midlands 5.02, West Midlands 6.11, East of England 6.40, London 8.87, South East 9.46, South West 5.80. Using private-rented-sector household counts instead of population moves the April 2026 representative price-to-rent ratio from 20.7 to 20.6 and no headline result materially.

Mortgage rate

Series CFMBJ95 (Bank of England Bankstats table G1.4): monthly average of UK monetary financial institutions' sterling weighted-average interest rate on loans secured on dwellings, new advances to households, January 2004 onwards, no missing months in the sample window.

Equity returns

MSCI publishes end-of-month ACWI index levels directly in GBP. I use the net-total-return variant (dividends reinvested after withholding tax) and convert levels to simple monthly returns. The baseline series deducts a fund fee of 0.12% per year, applied as a constant monthly drag: $r_t^{\text{net}} = (1+r_t)(1-0.0012)^{1/12}-1$. The FTSE 100 total-return comparator is the net-asset-value total return of a low-cost FTSE 100 ETF (dividends reinvested at NAV; ongoing charge 0.40%

per year to June 2014, 0.07% thereafter), which tracks the FTSE 100 total-return index with correlation 0.9999 over the months both are available.

Replication

All raw data under the Open Government Licence, all cleaning and analysis code, and scripts that re-download the licence-restricted financial series from their public endpoints are available at https://github.com/Martin-Cimprich/buy-vs-rent-england.

B Stamp duty schedules

Table 9 summarises the SDLT liability of a first-time buyer in England by purchase month, as implemented in the simulation. Before 4 December 2014 SDLT was a "slab" tax (the rate applied to the whole price); thereafter a "slice" tax (marginal bands). Where first-time-buyer relief exists, the relief schedule applies below its price cap and the standard schedule above it.

Table 9: SDLT for an England first-time buyer, by period

Period	**Schedule applying to a first-time buyer**
to 16 Mar 2005	Slab: 0% to £60k; 1% to £250k; 3% to £500k; 4% above
17 Mar 2005–22 Mar 2006	Slab, nil-rate band £120k
23 Mar 2006–2 Sep 2008	Slab, nil-rate band £125k
3 Sep 2008–31 Dec 2009	Slab, nil-rate band £175k (holiday)
1 Jan 2010–24 Mar 2010	Slab, nil-rate band £125k
25 Mar 2010–24 Mar 2012	First-time-buyer relief: 0% to £250k; standard slab above
25 Mar 2012–3 Dec 2014	Slab, nil-rate band £125k (relief withdrawn; 7% top band from Mar 2012)
4 Dec 2014–21 Nov 2017	Slice: 0% to £125k; 2% to £250k; 5% to £925k; 10% to £1.5m; 12% above
22 Nov 2017–7 Jul 2020	FTB relief: 0% to £300k, 5% £300–500k (no relief above £500k)
8 Jul 2020–30 Jun 2021	Holiday: 0% to £500k for all buyers; slice above
1 Jul 2021–30 Sep 2021	Standard nil-rate £250k; FTB relief 0% to £300k, 5% to £500k
1 Oct 2021–22 Sep 2022	Standard slice (£125k nil); FTB relief 0% to £300k, 5% to £500k
23 Sep 2022–31 Mar 2025	FTB relief: 0% to £425k, 5% £425–625k (no relief above £625k)
1 Apr 2025–	FTB relief: 0% to £300k, 5% £300–500k (no relief above £500k)

Sources: HMRC SDLT rate tables 2003–2026; House of Commons Library briefings SN07050 and CBP-9814. The simulation evaluates the schedule at each month's representative (or regional) average price. Scotland (LBTT, from April 2015) and Wales (LTT, from April 2018) operate different schedules and are outside the scope of this paper.